

Unix/Linux Operating Systems – Resit Exam

Duration: 1 hour

Total Points: 20

Strictly no phones, documents, or internet use. Cheating = automatic zero.

Name: _____

Section A – Theory & Concepts (6 pts)

Q1. (2,5 pts)

For each of the following commands or keywords, briefly explain what it does and give an example:

- a) alias
- b) export
- c) ssh
- d) scp
- e) rsync

Q2. (1 pt)

What is the main role of the shell in Linux? How is it different from the kernel?

Q3. (1 pt)

What is the purpose of & when used after a command in the shell?

Explain how to:

- a) Run a command in the background
- b) List background jobs
- c) Resume a background job in the foreground

Q4. (2 pts)

What does the pipe operator | do in a command chain?

Then write a pipeline that:

- Lists all `.sh`` files in the current directory
- Sorts the result alphabetically
- Counts how many files

Section B – Practical & Scripting (14 pts)

Q7. (1 pt)

Write a command that creates a folder named `archives`, compresses all `.txt` files in the current directory into a `data.tar.gz` archive, and moves the archive into `archives/`.

Q8. (1 pt)

Explain what this command does:

```
ls -l /etc | grep "^d" | wc -l
```

Q9. (2 pts)

Write shell commands to:

- a) Create a user named `user`
- b) Create a group named `unix`
- c) Add `user` to the `unix` group
- d) Set folder `class` so only `trainee` can read and write to it

Q10. (3 pts)

Write a script `convert\_scores.sh` that:

- Takes 3 grades as arguments (e.g. `./convert\_scores.sh 12 15 18`)
- Computes the average (equal weights)
- Displays:
 - "Excellent" if ≥ 16
 - "Good" if ≥ 14
 - "Pass" if ≥ 10
 - "Fail" otherwise
- Prints both the average and the evaluation

Q11. (2 pts)

Create a script `vowel.sh` that:

- Asks the user to input one lowercase letter
- Uses a `case` structure to determine if it's a vowel or a consonant